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Department of Agriculture and Aquatic Science

AGROCLIMATOLGY SA410 --LECTURE NOTE'S -2026

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Environmental Temperature and Crop Production

Solar radiation is the main source of heat energy to the biosphere. Temperature is the intensity aspect of heat energy, and it is of paramount importance for organic life. Temperature governs the physical and chemical processes that in turn control biological reactions within plants. It controls the diffusion rate of gases and liquids within plants, and solubility of plant nutrient substances is dependent on temperature. As such, environmental temperature has a primary role in plant growth and its geographical distribution over the earth.

SOIL TEMPERATURE

Soil temperature is an important environmental factor in plant growth and distribution. In comparison to air temperature, the amplitude of variation in soil surface temperature is much more pronounced because of the varying characteristics and composition of soil.

Factors Affecting Soil Temperature

a. *Aspect and slope:* These factors are of great importance in determining soil temperature outside the tropics. In the Northern Hemisphere, a south-facing slope is always warmer than a north-facing slope or a level plain. The reverse is the case in the Southern Hemisphere. The difference in soil surface temperature exceeds the difference in air temperature.

b. *Tillage:* By loosening topsoil and creating mulch, tillage reduces the heat flow between the surface and subsoil. Because a mulched surface has a greater exposed area and the capillary connection with moist layers below is broken, cultivated soil has greater temperature amplitude than uncultivated soil. At noon, the air temperature 2.5 cm above the soil surface can be 5 to 10 C higher in cultivated soil as compared to uncultivated soil.

c. *Soil texture:* Because of lower heat capacity, sandy soils warm up and cool down more rapidly than clay soils; hence, they are at a higher temperature during the day and a lower temperature at night.

d. *Organic matter:* Organic matter reduces the heat capacity and thermal conductivity of soil, increases its water-holding capacity, and has a dark color which increases its solar radiation absorptivity. In humid climates, because of a large water content, peat and marsh are much cooler than mineral soils in spring and warmer in winter. However, when organic soils are dry, they become warmer than mineral soils in summer and cooler in winter.

Soil Temperature and Crop Germination

Soil temperatures influence the germination of seeds, the functional activity of the root system, the incidence of plant diseases, and the rate of plant growth. Living tissues of many temperate plants are killed when they are exposed to a surface temperature of about 50^0C. Excessively high soil temperatures are also harmful to roots and cause lesions on the stem. Extremely low temperatures are equally detrimental. Low temperatures impede the intake of nutrients. Soil moisture intake by plants stops when they are at a temperature of 1 ^{0}C. Root growth is generally more sensitive to temperature than that of aboveground plant parts, meaning that the range between maximum and minimum temperature for roots is less than for shoots and leaves.

In numerous cases soil temperature is more important than air temperature to plant growth. In Canada, sowing of agronomically important crops takes place during the early months of spring when temperatures are well below the optimum. This often results in reducing the rate and success of germination, slow, asynchronous seedling emergence, and poor stand establishment.

Several rice varieties do not emerge as long as the soil temperature is below 11 ^{0}C Germination of warm-season grasses is very poor during the winter season. Slower germination rates during cooler seasons require long periods of soil water availability at the surface to enable germination.

Germination of sunflower, maize, and soybean is very poor when day/night soil temperature is above 21/12°C and soil water content is too low during the first week after sowing.

In the tropics, high soil temperature causes degeneration of the tuber in potato. Optimum soil temperature for this crop is 17 C. Tuber formation is practically absent above a soil temperature of 29 ^{0}C. Preconditioning of potato seed under specific temperatures has an important impact on germination.

Seed stored at 27°C showed the best germination, while that stored at 45°C failed to germinate even after eight days of lowering the temperature

in the germination environment to 17°C (Pallais, 1995).

Impact of Soil Temperature on Plant Growth

After germination, soil temperature is important for the vegetative growth of crops. For each species, a favorable soil temperature is needed for ion and water uptake. The daytime soil temperature is more important than the night time temperature, because it is necessary to maintain a favorable internal crop water status to match the high evaporation rate.

Maize yield is closely related to soil temperature at planting. Some cultivars sown at soil temperatures above 30°C show reduced final seedling emergence. Soil temperature controls the rate of maize development while the meristem is underground. Increased soil temperature accelerates the rates of leaf tip appearance and full leaf expansion, enabling the crop to more rapidly attain maximum green leaf-area index. This enables a better synchrony between the time of peak radiation interception and peak radiation incidence. The extent to which soil temperature affects yield will therefore vary with sowing time and the latitude of the crop's location.

Tomato seed germination, plant growth, and fruit yields are governed by the prevailing soil temperature conditions. Germination is completely inhibited at low temperatures (up to 5°C) as well as high temperatures (40°C). Germination is highest at 25 to 30°C. At 10°C, plant growth is slow, almost no fruit formation occurs, and plants start to die off prematurely. At 18°C, the highest growth rates and earliest fruit formation are recorded.

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The root zone temperature significantly affects the quality and yield of sweet pepper. Growth is more inhibited by low temperature than high temperature. Sugar content is influenced by root zone temperature. Phosphate content is lower at 13 and 33°C root zone temperatures than at other temperatures. Higher numbers of fruits are obtained at 18 to 28°C, and higher yields are obtained at 23 to 28°C than at other root zone temperatures. A 23°C root zone temperature is considered optimal for economic production of sweet pepper.

Optimal soil temperature for growth of wheat plant roots during the vegetative stage is below 20°C and is lower than that for the shoots. Temperatures higher than 35°C have been shown to reduce terminal root growth and accelerate its senescence. Root growth may cease altogether if soil temperatures drop below 2°C. Studies have shown that an air temperature of –20°C is lethal for root survival, although this must be translated into a soil surface temperature, which would, in most cases, be higher.

Cardinal Temperatures

Three temperatures of vital plant activity have been recognized, which are often termed *cardinal points.*

a. A minimum temperature below which no growth occurs: For typical cool-season crops, it ranges between 0 and 5 C, and for hot-season crops between 15 and 18 C.

b. An optimum temperature at which maximum plant growth occurs: For cool-season crops, it ranges between 25 and 31 C, and for hot-season crops between 31 and 37 C.
c. A maximum temperature above which the plant growth stops: For cool-season crops, it ranges between 31 and 37 C, and for hot-season crops between 44 and 50 C.

AIR TEMPERATURE

Air temperature is the most important climatic variable that affects plant life. The growth of higher plants is restricted to a temperature between 0 and 60 C, and crop plants are further restricted to a narrower range of 10 to 40 C. However, each species and variety of plants and each age group of plants has its own upper and lower temperature limits. Beyond these limits, a plant becomes considerably damaged and may even be killed. It is therefore the amplitude of variations in temperature, rather than its mean value, that is more important to plant growth. The midday high temperature increases the saturation deficit of plants. It accelerates photosynthesis and ripening of fruits. The maximum production of dry matter occurs when the temperature ranges between 20 and 30 C, provided moisture is not a limiting factor.

When high temperature occurs in combination with high humidity, it favors the development of many plant diseases. High temperature also affects plant metabolism. High night temperature increases respiration. It favors the growth of the shoot and leaves at the cost of roots, stolons, cambium, and fruits. It governs the distribution of photosynthates among the different organs of the plants, favoring those which are generally not useful for human consumption. High night temperature also affects plant metabolism. It accelerates the development of noncryophytes.

Most crop plants are injured and many are killed when the night temperature is very low. Tender leaves and flowers are very sensitive to low temperature and frost. Plants that are rapidly growing and flowering are easily killed. Low temperature interferes with the respiration of plants. If low temperature coincides with wet soil, it results in the accumulation of harmful products in the plant cells. Frost also interferes with plant metabolism. Spring wheat grain yields generally decline as temperature increases. Temperature stress intensity is severe under late sowing, causing a reduction in the duration of later growth phases. Grain test weight, spikelets/spike, and grains/spike under hot (normal sowing) environments and spike length and spikes/m2 under very hot (late sowing) environments are adversely affected.

Other factors that significantly contribute to yield under high temperatures are tiller numbers and reduced height.

Grain yield of rice is highly correlated with minimum temperature. A prediction model in the Philippines showed that the high yield observed especially during the dry season is due to lower minimum temperature. Higher grain yield corresponds with a seasonal minimum temperature of 22.5°C, compared to an average seasonal minimum temperature of 24.2°C.

TABLE 3.1. Cardinal temperatures for the germination of some important crops

	Cardinal temperature (°C)		
Plant	**Minimum**	**Optimum**	**Maximum**
Wheat	3-4.5	25	30-32
Barley	3-4.5	20	38-40
Maize	8-10	32-35	40-44
Rice	10-12	30-32	36-38
Tobacco	13-14	28	35
Sugar beet	4-5	25	28-30
Peas	1-2	30	35
Oats	3-4	25	30
Sorghum	8-10	32-35	40
Lentils	4-5	30	36
Carrot	4-5	8	25
Pumpkin	12	32-34	40

Reasons for low and variable cotton yields are associated with extremes of temperatures. Yield and fiber characteristics respond to variations of daily mean and amplitude of temperature.

Mean temperature reduction improves yield components, but high temperatures, particularly high day temperatures, increase fiber length, uniformity, and strength. Large daily temperature amplitude produces an intermediate number of flowers and the lowest retention percentage. Fruiting and yield are increased by a reduction in temperature down to the threshold mean temperature of 22°C. An adverse effect of low minimum temperature on lint and fiber properties was also observed.

Sowing date, reflecting temperature conditions, significantly affected phenology (time to emergence, flowering, and maturity) and pod yield of groundnut. The observed responses appear to have been due to the effect of temperature differences on partitioning during the pod-filling phase.

Temperature and Photosynthesis

The rate of photosynthesis and respiration increases with an increase in temperature, until a maximum value of photosynthesis is reached. This value is maintained over a broad range of temperatures. Then, at considerably higher temperatures, when the enzyme becomes inactivated and various reactions are disturbed, photosynthesis decreases and ultimately stops. The range of temperature in which photosynthesis is more than 90 percent of the maximum obtainable can be regarded as optimum. This range is narrower for net photosynthesis than for gross photosynthesis, because while gross photosynthesis is still operating at top speed in the optimum

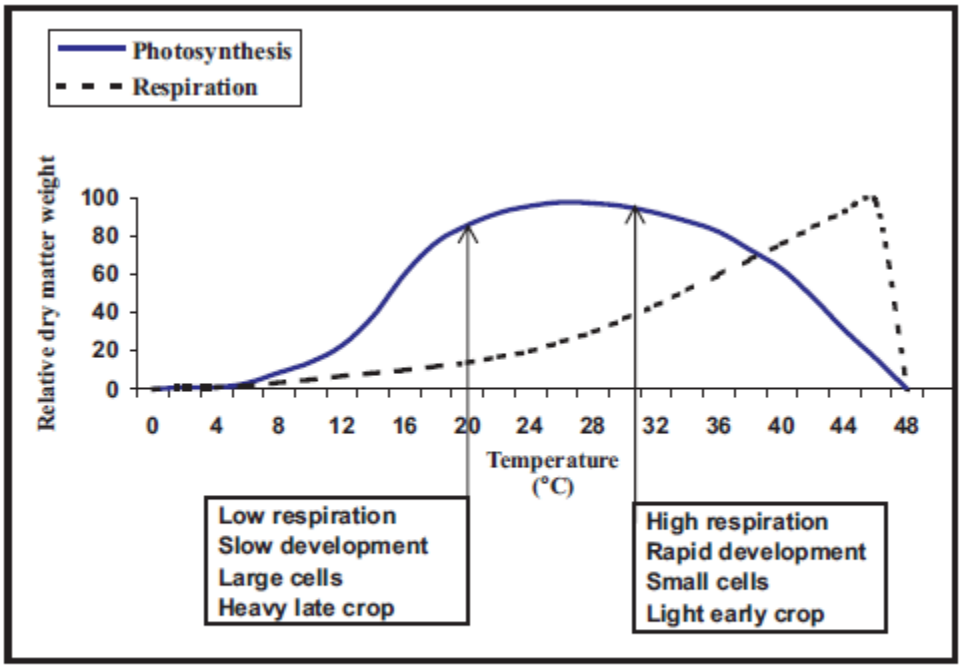


FIGURE 3.2. Effect of temperature on photosynthesis and respiration of potato (*Source:* Mavi, 1994.)

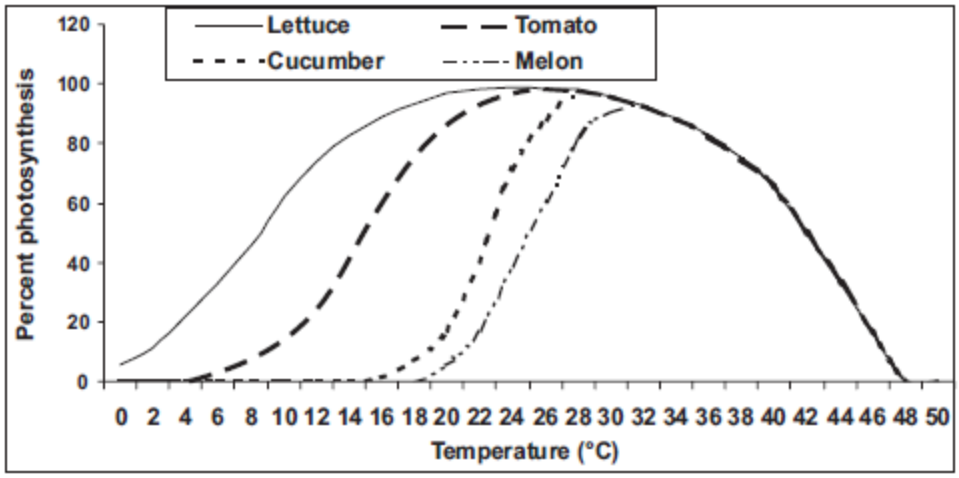


FIGURE 3.3. Temperature limits for photosynthesis (*Source:* Mavi, 1994.)

range of temperatures, the rate of respiration increases, diminishing the net photosynthetic yield. The temperature limits for net photosynthesis for some plant groups. .

PLANT INJURY DUE TO SUDDEN CHANGES IN TEMPERATURE

Living organisms receive and transfer thermal energy through radiation, conduction, and convection. Transpiring water to the surrounding atmosphere also transfers thermal energy from growing plants. Through these processes, they remain in equilibrium with the surrounding environment and maintain normal growth and development. However, with the passage of weather systems, changes in atmospheric temperature are often very sudden, and plants cannot adjust to these severe variations and are damaged beyond recovery.

Leaf Temperature versus Air Temperature

Under normal conditions, leaf temperature remains around the ambient temperature but differs under certain situations. At a temperature of about 33 C, there is a tendency for equality between air and leaf temperature. Below this temperature, leaves tend to be warmer than the air and vice versa.

TABLE 3.2. Temperature dependence of net photosynthesis during the growing season under conditions of natural carbon dioxide availability and light saturation

	Temperature limits (°C) for carbon dioxide intake		
Plant group	**Lower limit**	**Optimum**	**Upper limit**
C4 plants of hot habitats	–5 to 7	35 to 45	50 to 60
C3 crop plants	–2 to over 0	20 to 30	40 to 50
Alpine plants	–7 to –2	10 to 20	30 to 40
Evergreen tropical trees	0 to 5	25 to 30	45 to 50
Deciduous trees of temperate zone	–3 to –1	15 to 25	40 to 45
Evergreen conifers	–5 to –3	10 to 25	35 to 42
Shrubs of tundra	–8	15 to 25	40 to 45
Lichens of cold regions	–25 to –10	5 to 15	20 to 30

Where the temperature exceeds 33 C, leaves appear to suffer from water stress.

- *Leaves exposed to sun:* Thick leaves that are not transpiring actively in still air are several degrees warmer than the air. Under intense radiation and high humidity, some leaves may be at a temperature 15 C higher than the air. Likewise, under very hot and low humidity conditions, leaf temperature can be as much as 10 C higher than the air. Where plants do not suffer for want of moisture, the difference between leaf and air temperature is very small.
- *Leaves under shade:* Leaves shaded from direct sunlight are usually somewhat warmer than the surrounding air.
- *Leaves exposed to a clear night:* At night when the sky is clear, leaf temperature is usually lower than the air temperature. During a cold and clear night, a leaf may be around 2 C cooler than the surrounding air.
- *Leaves exposed to a cloudy night:*With cloud cover, the difference in air and leaf temperature is small. In certain cases the leaf temperature may be slightly higher than air temperature.

High-Temperature Injury to Plants

Thermal death point of active cells ranges from 50 to 60 C for most plant species, but it varies with the species, the age of tissue, and the length of time of exposure to high temperature.

For aquatic and shade plants the lethal limit is 40 C, and for most xerophytes it is 50 C, when the plants are exposed to a saturated atmosphere for about half an hour. High temperature results in the desiccation of the plant and disturbs the balance between photosynthesis and respiration.

Once the temperature exceeds the maximum up to which growth takes place, plants enter a state of quiescence. When the temperature becomes extremely high, a lethal level is reached. At temperatures higher than the optimum cardinal, the physiological activity declines as a consequence of inactivation of enzymes and other proteins. Leaf functions are disturbed at about 42 C, and lethal effects on active shoot tissues generally occur in the range of 50 to 60 C.

Many rice varieties subjected to high temperature just before and just after flowering result in more than 20 percent sterility. High temperature just before or during flowering decreases pollen size, causes a shortage of starch in pollen, and increases the proportion of anthers that did not dehisce. High temperature during ripening decreases grain weight. In wheat crops, a major effect of high temperature appears to be the acceleration of senescence, including cessation of vegetative and reproductive growth, deterioration of photosynthetic activities, and degradation of proteinaceous constituents.

Serious damage to fruit and vegetable crops resulting from excessively high temperature has also been recorded.

Apart from desiccation and disturbed photosynthesis and respiration balance, plants are injured in several ways, such as excessive respiration from seeds, sun scald, and stem girdle.

The higher the temperature, the greater is the rate of respiration, which results in the rapid exhaustion of food reserves of seeds. Temperatures on the sunny side of the bark on stems during hot afternoon and late night undergo great fluctuations. The injury inflicted because of this short period fluctuation in temperature is known as *sun scald.* Stem girdle is another injury associated with high temperature. Exceptionally high temperature at the soil surface and the adjoining laminar sublayer of the air frequently scorches the short stems. The scorching of the stem is known as *stem girdle.*

This type of injury is most common in young seedlings of cotton in sandy soils where the temperature of the soil surface during summer afternoons may be as high as 60 to 65 C.

Stem girdle injury is first noted through a discolored band a few millimeters wide. This is followed by shrinkage of the discolored tissues. It appears that stem girdle causes the death of

plants by destroying the conductive tissues or by an injury that helps the establishment of pathogens.

Low-Temperature Injury to Plants

Exposure to extremely low temperatures and heavy snowfall damages the plant in several ways including suffocation, desiccation, heaving, chilling, and freezing.

Suffocation

Small plants may suffer from deficient oxygen when covered with densely packed snow. When suffocated, certain toxic substances accumulate in contact with roots and crowns and tend to inhibit the diffusion of carbon dioxide.

Physiological Drought and Desiccation

Spring drought sometimes occurs in coniferous trees in cool temperate climates. This results from excessive transpiration and a time lag in absorption of moisture from the soil, caused by a warm period when the soil is still frozen. The result is an internal moisture deficit sufficient to cause death of the twigs. The decreased water absorption by plants at low temperatures is the combined effect of the decreased permeability of the root membrane and increased viscosity of water. This results in increased resistance to water movement across the living cells of the roots.

Heaving

Injury to a plant is caused by the soil layer lifting upward from the normal position and causing the root to stretch or break at a time when the plant is growing. Sometimes the roots are pushed completely above the soil surface.

After thawing, it is difficult for the roots to become firmly established, and the plants may die because of this mechanical damage and desiccation.

Chilling

Plants of tropical origin are damaged by exposure to mild chilling for two to three days. Plants of temperate origin withstand chilling for long periods without suffering any injury. Rice, cotton, and cowpea are killed when exposed to temperatures near 0 C for about two to three days. Sudan grass and peanuts are injured by short exposure to chilling temperatures but recover if favorable temperature conditions return shortly afterward. Short duration mild chilling does not seriously injure corn, sorghum, and pumpkin plants.

Plants of cool climate origin such as wheat and soybean are injured when exposed to prolonged chilling but recover with the return of favorable conditions.

In temperate climates, two types of injuries occur because of low temperature. These are delayed growth and sterility. For example, rice yield decreases due to insufficient grain maturation caused by low temperatures during the ripening period. When flowering is delayed by low temperatures at a certain stage before heading, insufficient time is available to the grains to ripen fully before frost occurs in autumn.

In the sterility type of injury, rice yields decrease due to sterile spikelets caused by low temperatures at the booting stage or at anthesis. The observed injury in developmental order is a stoppage of anther development; pollen unripeness; partial or no dehiscence; pollen grains remaining in anther loculi; little or no shedding of pollen grains on the stigma; and failure of germination on the stigma.

Chilling injury of fruits is of particular interest because they are often stored and shipped at low temperatures. Symptoms of chilling injury to fruits include surface pitting, lesions, discoloration, susceptibility to decay organisms, and shortening of storage life. Fruits subjected to chilling injury do not ripen normally. The critical temperature at which chilling injury occurs is 8 to 12 C for tropical fruits such as banana, avocado, and mango, and 0 to 4 C for temperate zone fruits such as apple.

Freezing Injury

Plant parts or an entire plant may be killed or damaged beyond repair as a result of actual freezing of tissues. Freezing damage is caused by the formation of ice crystals, first in the intercellular spaces and then within the cells. Ice within the cells causes more injury by mechanical damage disrupting the structure of the protoplasm and plasma membrane. Freezing of water in intercellular spaces results in withdrawal of water from the cell sap, and increasing dehydration causes the cell to die.

FROST: DAMAGE AND CONTROL

Frost is a climatic hazard that causes serious damage to standing crops in temperate and subtropical climates. In Australia, crop losses due to frost are huge.

Frost is a weather hazard that occurs when the environmental temperature drops below the freezing point of water. It can be a white frost (also known as hoarfrost) or a black frost. White frost occurs when atmospheric moisture freezes in small crystals on solid surfaces. Black frost

occurs when few or no ice crystals are formed because air in the lower atmosphere is too dry, but the damaging effect of the low temperatures on vegetation is the same as that of white frost.

Frost is formed through the physical processes of radiation and advective cooling. These are referred to as radiation frost and advection frost, respectively

• *Radiation frost* occurs when a clear sky and calm atmosphere (winds less than 8 kph) allow an inversion to develop, and temperatures near the surface drop below freezing. The thickness of the inversion layer varies from 10 to 50 m. The term inversion comes from atmospheric conditions being inverse to the normal daytime condition in which air temperature decreases with height. Plants can be successfully protected from radiation frost.

• *Advection frost* occurs when a cold air mass invades a relatively warm area suddenly. Under advection frost, winds may be above 7 kph and clouds may exist. The advected cold air mass may be 150 to 1,500 m deep. Plants can be protected from advection frost to a limited extent.

Frost Damage to Plants

Damage to plants from frost occurs because it results in freezing of the plant tissues. Freezing of plant tissues is a physical process triggered by icenucleating bacteria, the intensity and duration of the night temperature to which the plants are exposed, and the plant growth stage

THERMOPERIODISM

The response of living organisms to regular changes in temperatures, either day/night or seasonal, is known as *thermoperiodism.* Thermoperiodism exerts effects on the seasonal biology of insects and the growth and development of plants. Effects on insects include rates of growth and development, determination of diapause and dormancy, and acclimatization to low temperatures. Effects on growth and development of plants vary from one species to another. Crops such as soybean, maize, tomato, potato, eucalyptus, and mango are classified as thermoperiodic, while wheat, oats, peas, and cucumber are classified as nonthermoperiodic.

In soybean, a cool day/night temperature combination of 18/14°C disrupts floral development, leading to physically malformed parts. Normal floral initiation and pod development occurs at 30/18°C and 30/22°C, respectively, while the greatest number of pods per plant is obtained at 26/14°C. Tomatoes grow faster when the temperature is 26°C by day and 17°C by night than at the constant temperature of 26°C or any intermediate temperature. For this reason, tomatoes do not grow well in warm countries except in those locations where the temperature falls appreciably at night.

Some maize cultivars respond to a daily temperature fluctuation. In experiments with three lines differing in earliness (early, mid, and late) grown in a phytotron from two-week-old seedlings raised in a 16 h photoperiod under three different day/night temperature regimes, there was a direct correlation between chlorophyll content and grain yield. It proved best for yield and energy savings to raise seedlings of early lines at a high diurnal temperature (day/night = 24/15°C), and midseason or late lines at a constant day and night temperature of 21°C .

Potato plants grown under the fluctuating temperature treatment develop normally, develop tubers, and have a fivefold or greater total dry weight compared to those under constant temperature. This suggests a thermo period could allow normal plant growth and tuberization in potato cultivars that are unable to develop effectively under continuous radiation.

Two important aspects of the environment influencing induction of flowering in mango are photoperiodism and thermoperiodism. Studies in the indicates that minimum temperature below 10°C and above the freezing point stimulated heavy flowering in mango. Furthermore, flowering occurred only in a single flush, compared to two to three flushes under normal environmental conditions.

Many crop seedlings will grow perfectly well at a constant temperature, but others, such as celery, germinate best at fluctuating temperatures. The emergence of carrot seedlings from soil is faster in fluctuating temperatures than at constant temperature. *Solanum elaeagnifolium* is a weed that produces solasodine (a steroidal alkaloid used for the production of corticosteroids).

It has a strict requirement for alternating temperature to germinate. A constant day/night temperature prevents the germination process. Seeds become sensitive to alternating temperature five days after the start of imbibition. After that, three daily cycles of alternating temperature are required for 50 percent germination.

The ecological significance of this response to a diurnal alteration of temperature may be that it promotes germination of those seeds close to the soil surface. When such fluctuations do not occur, germination, especially of more deeply seated seeds, may remain suppressed.

Apart from daily fluctuations in temperature, seasonal fluctuations are important in the development of many plants. Annual plants do not need a cold period during their development, except for plants that germinate in autumn and flower in the spring or summer after a cold winter. An example is winter wheat. Peaches cannot flower at high temperatures, but the

vegetative growth phase continues. They need a period of cold weather before flower buds can open. No flower primordia are laid down under conditions of continued high temperature.

ROLE OF WEATHER AND CLIMATE ON CROP PESTS AND PARASITES IN ANIMALS

Weather is the most important factor that determines the geographical distribution and periodic abundance of crop insect pests and parasites of animals. Weather controls the development rate, survival, fitness, and level of activity of individual insects; the phenology, distribution, size, and continuity of insect populations; migration and their establishment; and the initiation of insect outbreaks. Weather influence can be immediate, cumulative, direct, indirect, time lagged, exported, or imported. Indirect effects arise through host quality and parasite populations. A time-lagged influence is one that occurs at a later stage as a consequence of both past and current weather. Imported/exported influences arise because insects are highly mobile, and outbreaks may be initiated by windborne migrations.

Among the weather elements, temperature, humidity, and wind play the major roles in insect life. Other elements of lesser importance are solar radiation and photoperiod. In interpreting the role of weather in an insect's life, we must remember that all weather elements are interrelated, so the role played by any individual element is not simple to understand and explain.

Temperature

Each species of insect has a range of temperature within which it can survive. This range is referred as the tolerable zone. Within the tolerable zone, there are different optimal temperature ranges for a variety of vital functions. Exposure to a temperature toward the upper or lower limit of the tolerable zone will usually result in death if it persists for a long enough time. At extremes of the tolerable zone, death will occur after a short duration of exposure.

The actual temperatures that limit the tolerable zone vary from species to species, but extremes exist that apply to all insects. Most insects have an upper temperature tolerance between 40 and 50°C, and no known insect can survive temperatures in excess of 63°C.

Some insects can adapt physiologically to survive several months of hot, dry weather in a dormant state called summer diapause. The absolute minimum temperature tolerated by any insect is not well defined but is almost certainly below –30°C. Some insects cannot survive for

long if the temperature drops below the lower threshold for development. Other species can become dormant at low temperatures. Activity and development cease but begin again as soon as the temperature exceeds the activity or developmental thresholds. Yet others usually overwinter in a particular stage that is physiologically adapted and therefore can survive periods of extreme cold. Several insects inhabiting temperate or even arctic regions are able to survive by the process of supercooling, in which tissues are able to withstand the freezing of their fluid for extended periods without damage. The mean supercooling point for larvae of elm bark beetle *Scolytus laevis* reached as low as –29°C in midwinter (Hansen and Somme, 1994).

The duration of the time of exposure to extremes of temperature is also important to the survival of insects. The pecan aphid, *Monelliopsis pecanis* (Hemiptera: Aphididae), could survive fairly well even at temperatures near freezing if exposed for only one hour, but many deaths occurred if they were forced to spend five hours at these low temperatures. At very high temperatures, survival is reduced even after a one-hour exposure.

Insects are able to function faster and more efficiently at higher temperatures. They can feed, develop, reproduce, and disperse when the climate is warm, though they may live for a shorter time. Higher temperatures are not always favorable to insects, usually reducing their life span. All other factors remaining normal, insects live for shorter times at higher temperatures. An example is that of the parasitic wasp *Meteorus trachynotus* (Hymenoptera: Braconidae). Variation of temperature from 15 to 30°C reduces its adult life span from 40 days to a mere 10 days or so.

Low temperature is an advantage under certain conditions. For example, there are lower energy demands at low temperatures. If resources such as food are in short supply, insects can survive longer without starving. Under extremely low temperatures some insects can remain in suspended animation until warmer conditions return.

In temperate regions, where the seasonal variation in temperature is often large, development starts slowly in early spring, progresses more rapidly as the season advances, and perhaps slows or is suspended in the heat of midsummer. Temperature is less limiting in the tropics and subtropics, but arrested development is commonly used to survive a dry season. Survival through unfavorable seasons is usually possible only in particular developmental stages.

Temperature also impacts insect activity. As the temperature rises, insects move more rapidly. Short-distance flight activity in relation to temperature has been documented for a number of insect species in Australia.

Major meteorological factors affecting migration are the vertical profiles of temperature, wind velocity, and the presence of convergence. Temperature may determine the time of flight, height of flight, and thus the speed and direction of the transporting wind, as well as flight duration. *Chortoicetes terminifera* adults initiate their long-distance nocturnal flights only if the temperature at dusk (the time of takeoff) exceeds 20°C and rain has fallen on them at certain stages during their development.

The flight duration and dispersal distance of *E. postvittana* adults is much greater when the larvae are reared at high (25-28°C) rather than low (15°C).

Flight duration may sometimes be determined by the migrants' supplies of physiological fuel, which could depend on food abundance and quality (and thus in turn on weather conditions) at the development stage. Direct influences leading to the termination of migration include falling temperatures below the threshold for flight, the onset or cessation of thermal convection, and an encounter with rainfall that can cause flying insects to be washed down.

Temperature also affects behavior of insects. Insects may remain totally inactive at both high and low temperatures or move actively along a temperature gradient until a preferred zone is encountered. The influence of temperature on dispersal, mating, and reproduction is of great importance. If conditions are adverse for dispersal, local populations neither increase because of influx, nor decline because of exodus. If temperatures are not suitable for mating for several days, some adults may die without leaving offspring and others may become less fertile due to age.

Moisture

The moisture content in the habitat of an insect directly determines whether or not an individual survives. Moisture also has indirect effects on insect populations through its influence on plant growth. All forms of environmental moisture (atmospheric humidity, rain, snow, hail, dew, soil moisture, and surface water) influence the water balance of insects.

The humidity in an insect's habitat may have some indirect effects. Some parasites do not search for hosts or oviposit in them if the relative humidity is either low or high. The susceptibility of insects to fungal, bacterial, and viral diseases also changes with environmental moisture. Moist

conditions seem to facilitate the spread of some insect pathogens and may also affect their survival and virulence.

Rainfall can act as a direct cause of mortality. Insect eggs and small larvae can be permanently washed from their host plants by heavy rain. Rain may also saturate the soil and drown insects that are unable to escape. Many insects cease feeding during periods of precipitation and may seek refuges in which to pass a rainy period. Small parasitoids have difficulty moving around in wet conditions. A prolonged rainy spell, particularly when the temperature is suitable for development, may lengthen the time required to complete development or cause mortality by starvation.

Heavy or excessive rain can cause high mortality, either directly through knockdown, saturation, or flooding, or by providing conditions favorable for disease. Heavy rain washes aphids off of their host plants, and both beetles and bugs may be killed by violent thunderstorms. Insect abundance varies with seasonal variations in rainfall. Some species are more abundant in the dry season, whereas others proliferate only during the rainy season. Lack of rain can cause desiccation and death of insects.

Rain may not have an influence at the time it falls but may promote insect performance some months later. This phenomenon is well illustrated by the seasonal outbreaks of African armyworm, *Spodoptera exempta* (Lepidoptera: Noctuidae) which can reach enormous numbers to become serious pests of cereals and pasture. In Kenya, the number of outbreaks was negatively correlated with rainfall in the six to eight months preceding the start of the armyworm season. The high correlation between rain and later armyworm outbreaks has been used to construct a prediction model for Kenya, providing an accurate forecast of the likelihood of armyworm outbreaks (Haggis, 1996).

Rain also plays a role in altering a host's susceptibility to windborne insects and disease vectors. Rain-drenched or moisture-stressed crops and stock may be particularly vulnerable to insects or the pathogens they carry (Risch, 1987). Sheep are especially susceptible to strike by *L. cuprina* during periods of frequent rainfall, which increase the incidence of predisposing factors such as fleece rot and nematode infestation.

In the region where the vector overwinters, the disease is endemic, and cows usually develop immunity before becoming pregnant (Murray, 1987). Jarrah trees stressed by low rainfall during the previous winter are particularly susceptible to attack by *Perthida glyphopa* (Mazanec, 1989).

Wind

Wind is an important factor of the environment of insects, and it influences insect populations in a number of ways. It is a vital component of broad weather patterns, giving rise to fronts and convergence zones. Low pressure systems and anticyclones in temperate regions determine migration trajectories of insects, while trade winds and monsoons determine the trajectories in tropical and subtropical areas. Wind causes insect displacement and therefore affects population changes by influencing the numbers moving into or out of an area. It can carry them considerable distances away to new habitats and regions. Many insects and pathogens appear to undertake enormous migrations covering hundreds if not thousands of kilometers on occasions. They perform this feat by exploiting the wind as an external source of energy.

Light

Light is not a true climatic factor, but it is interrelated with solar radiation and temperature. As such, it is often included as a component of weather. Photoperiodism (variations in the photoperiod in a 24-hour day-night cycle) exercises a great deal of control over processes directly related to survival of insects. Light intensity greatly influences insect behavior. Many species are active during the hours of full sunlight but remain quiescent at night. Some insects are active during the faint light of dawn and dusk.

Day length can also be used as a signal or trigger by insects to enter diapause during potentially harsh conditions such as summer heat, winter cold, and drought. In some cases, the day length experienced by an insect larva provides information about the progression of the seasons. The ability to vary growth and development rates enables the insect to achieve efficient timing relative to favorable conditions (Leimar, 1996).

SOME IMPORTANT INSECT PESTS OF CROP PLANTS

Aphids

Weather factors, especially temperature and rainfall, play a dominant role in the population dynamics of aphids in all the climatic regions of the world where crop production is possible. Aphids are highly sensitive to temperature changes. Field observations, climate chamber experiments, and computer simulations confirm this fact. Aphids in the tropics show remarkable adaptability to climate, regulating their population by suitably adjusting their life history strategy development, reproduction, survival, and dispersal.

Armyworms

Economic infestations of armyworms in many parts of the world have revealed that precipitation is the primary factor influencing pest populations. However, temperature and the availability of weather transport systems were the most important climatic factors governing pest abundance.

Grasshoppers

Temperature sensitivity determines the geographical distribution of grasshoppers, with generalist species widespread and thermally specialized species restricted to warmer habitats. For all the species that are thermal specialists, variation in their sensitivity to temperature is a good predictor of their distribution. The developmental and reproductive responses to different rearing temperatures of grasshoppers, examined in a laboratory experiment, showed that growth and development rates increase with temperature for each species. Nymphal development, adult mass and size, and egg production rate also increases with temperature.

Locusts

Locusts are able to travel long distances and colonize new habitats. Therefore, their distribution is variable in time and space and can occur within a large area. The pest is feared for both its destructive capacity and its constant threat to the region. It is capable of sudden appearances and severe devastation to standing crops. The most affected part of the world is the continent of Africa, where in some years the infested area may cover several million hectares. Distribution and sequence of rainfall is the principal determinant of locust population increase over several generations and of the concentration of widespread populations into highly mobile and destructive swarms.

Survival and populations are greatest with an increased frequency of sufficient rainfall, where rainwater is enhanced by runoff and flooding, and where vegetation and soils provide suitable habitats. However, excessive rainfall affects the first three nymphal stages, limiting its development.

Cotton Bollworms

The effect of weather conditions on various species of bollworms has been investigated in many cotton-growing regions of the world in laboratory as well as in field. The combined effect of weather factors, maximum temperature, minimum temperature, rainfall, and relative humidity on the population density of cotton bollworms is very high. Laboratory studies have demonstrated the temperature-dependent development of larvae and pupae of pink bollworm, *Pectinophora*

gossypiella (Saund). No eggs hatched at less than 10° or more than 37.5°C. Mortality of larvae and pupae also increased at terperatures greater than 37.5°C. The development rate of all stages of the pest increased with temperature.

Development of larvae was successful at all temperatures between 15° and 35°C. Larval period and adult longevity decreased as relative humidity increased. The most important weather factor for the abundance of cotton bollworm is the amount and distribution of rainfall. The incidence of cotton bollworms (Lepidoptera) in southern China is significantly affected by July and August rainfall. Populations of the pests were sparse when total precipitation was greater than 500 mm and dense when total precipitation was less than 400 mm, showing a very significant negative correlation. Continuous rain produced more severe damaging effects on the pupal stage (Li et al., 1996).

Fruit Fly

The abundance of fruit fly is greater in regions where the daily maximum temperature does not exceed 38°C during summer. Immature adults are unlikely to survive when the maximum temperature exceeds 40°C for four continuous days. However, adults are mobile and can seek cooler habitats in the field and survive very hot weather periods when the temperatures suggest that they should not (Meats, 1981). Conversely, the lowest monthly mean of 2°C is only marginally favorable for winter survival.

Moisture appears to be the primary determinant of the number of fruit flies, and correlations between summer rainfall and fruit fly populations are significant. Populations reach extremely high numbers in wet years and decline in dry years. Other factors being favorable, a relatively high population would survive with a mean monthly rainfall of 48 mm and more. Mavi and Dominiak (2000) observed a highly significant correlation between the availability of moisture, as measured by summer rainfall, and the peak numbers achieved each year. Rainfall in excess of 170 mm in November, January, and February resulted in high fruit fly populations, and less than 170 mm resulted in low populations. Meats (1981) reported that more than 48 mm rainfall per month in summer resulted in at least three generations.

CLIMATE AND PARASITES OF ANIMALS

Worms, flies, lice, and ticks are the major parasites of cattle, sheep, pigs, and poultry. Diseases caused or carried by parasites constitute a major obstacle to the development of profitable livestock enterprises. Some of these parasites also have significance for human health.

Parasites and many parasitic diseases are influenced by climate. Diseases spread by insects are encouraged when climatic conditions, temperature and excessive moisture in particular, favor the propagation of the vector. Many parasites and parasitic diseases in cattle and sheep reach their peak incidence in warm, wet summers and are relatively rare in dry seasons. Internal parasites are similarly influenced by climate. The direction of prevailing winds is of importance in many disease outbreaks, particularly in relation to the contamination of pasture and drinking water by fumes from factories and mines and the spread of diseases carried by insects.

HELMINTH PARASITES

The helminth parasites of sheep and cattle are classified into nematodes, trematodes, and cestodes. In everyday language these are roundworms, flukes, and tapeworms. Climate is the major factor determining the development of free-living parasites, and it affects different species of parasites differently. It also affects the survival of the parasites; for example, as temperature varies from warm to cold, and levels of soil and plant moisture vary from moist to dry, populations of the free-living stages of parasites fluctuate. The differential effects of climate for different species of nematodes explains why haemonchosis is the common parasitic disease in summer-rainfall areas and ostertagiosis is the common one in winter-rainfall areas.

Most helminth parasites undergo a period of development outside the host before becoming infective for another host. During this time in the outside world, the rate of their development and chances of survival are influenced by climatic and other environmental factors. The rate of development of the extra-host stages tends to rise with increasing temperature. Moisture is required for survival, and extreme desiccation is usually lethal. In the wetter parts of the tropics, extra-host stages are seldom exposed for long to the destructive effects of desiccation, and temperatures are commonly optimal throughout the year. Under these conditions, the survival rate of parasitic forms outside the host is high; they develop rapidly to the infective stage and large populations are established.

Even where prolonged dry seasons alternate with very short wet seasons, the extra-host stages of many endoparasites can take full advantage of the warm, wet conditions associated with the

latter, while the adult stages which subsequently develop within the host are well protected and safely carried over to the ensuing dry season.

Nematodes

Roundworms or nematodes generally cause major trouble to livestock in tropical areas. The level of parasitism in grazing animals depends to a large extent on the numbers of free-living stages on the pasture. Climate and pasture management affect the survival of these free-living stages of gastrointestinal nematodes. Roundworms or nematodes require oxygen, warmth, and moisture for their development and completion of their full life cycles. They are susceptible to dehydration. In areas where the dry season is prolonged and severe, pastures are often free of infection long before the beginning of the rains. In such areas worm infection is carried over from one wet season to the next by infected carrier animals.

Flukes

Fluke or trematode infections are found in all species of domestic animals, including poultry, but cattle and sheep are the principal victims. Flukes are prevalent in most animal and sheep grazing areas of the world. Fluke commonly use a water snail as an intermediate host. Infection with this species is associated with stock grazing in land that may be infested by migrating snails. The level of infestation with liver fluke is often heaviest following rain, flooding, and irrigation, though in areas of limited infestation, such as banks of streams or dams, it is heaviest in dry times when animals graze these areas more closely.

ARTHROPOD PARASITES

Arthropod parasites, such as flies, mosquitoes, and midges, assume economic significance by impairing productivity or carrying pathogens. Their greater importance is because they transmit infectious diseases.

Flies

Biting and bloodsucking flies affect cattle by sucking blood and irritating the stock. Because the larvae and pupae of these flies need water for their metamorphic life, they are more plentiful during the rainy season, as compared to the dry to the dry season. Blowfly strike *(Lucilia cuprina)* in sheep is primarily governed by climatic conditions. Flywaves that cause colossal mortalities of sheep occur mainly in spring and autumn when prolonged rainy spells keep the sheep wet to the skin.

The **tsetse fly** is notoriously important because of its role in the transmission of trypanosomiasis, or "nagana," of cattle and other animals in Africa. It also causes sleeping sickness in humans. Vast tracts of tropical Africa are infested with tsetse flies and are literally closed to successful animal husbandry because of trypanosomiasis.

There are many different species of tsetse fly, and various groups have particular biological requirements. In general, riverine species require the cool, moist environment associated with tree-lined riverbanks. On the other hand, savanna species inhabit dry, open parkland. The distribution of tsetse, as revealed by various surveys, is strongly influenced by environmental conditions such as climate and vegetation cover.

Mosquitoes

Mosquitoes are common and widespread throughout the world. They are important as vectors of Rift Valley fever (hepatitis enzootica) of sheep and other animals, African horse sickness, malaria, lymphatic filariasis, yellow fever, western equine encephalitis, and *Dirofilaria immitis* (a dog heartworm) (Speight, Hunter, andWatt, 1999). In semiarid tropical and subtropical regions, severe plagues usually occur after spring and summer rains. A close relationship exists between El Niño and seasonal moquito populations and mosquito-associated diseases.

Lice

Lice are small obligate parasites that are highly host specific. They are dorsoventrally flattened, wingless, and have tarsi adapted for clinging to hair or feathers. Lice infestations are common in cattle, sheep, goats, and horses in late winter and early spring. According to Arundel and Sutherland (1988), *Bovicola ovis* is the most common and important louse on sheep, causing considerable irritation that results in poor wool growth and damaged wool. Infested sheep rub against objects such as fences and fence posts and bite and chew their fleeces. The fleeces therefore appear deranged and have a pulled and ragged appearance, particularly the areas on the sides behind the shoulder, which sheep can reach with their mouths. Solar radiation, temperature, and rainfall all have profound effects on lice numbers. Lice is an important parasite of sheep and cannot survive off sheep for more than a few hours under extremes of temperature and light. Laboratory studies have, however, shown that lice held away from sheep could survive for 11 days at 25°C.

Ticks

Ticks occur on a global scale but they transmit many more serious viral and protozoan diseases in tropical and subtropical areas than in other regions. Climate and vegetation are the major factors affecting tick distribution When weather conditions are favorable, ticks lie in wait (on grass or rocks) or move in active search of a host. When conditions are unfavorable the ticks return to shelter (under a stone, in litter, under vegetation). Depending on the climatic characteristics of a season and region, ticks may be active during the day (morning or evening). Ticks are active at night in dry plains where strong sunlight prevents any diurnal movement on the ground. Each species of tick has a particular threshold temperature below which a diapause occurs in all instars. Various climatic factors such as sunlight, temperature, rainfall, and wind patterns condition the presence or absence of a tick species (Shah and Ralph, 1989).

Development and survival of free-living stages of tick are related to temperature, while the duration of survival is influenced mainly by rainfall and consequent relative humidity (Pegram and Banda, 1990). Appropriate relative humidity, rather than wet conditions, is essential for the development and survival of eggs and pupae and the survival of unfed hatched ticks. Each species of tick is adapted to a particular relative humidity range which varies with the instar and its size. The requirements range from extremely humid to very low relative humidity. Immature stages have very specific requirements, while adults can protect themselves better against evaporation because of their larger size and thicker tegument. Immatures adjust their humidity requirements by locating in holes in the ground, cracks in rocks, under litter, at the base of the vegetation layer, and in other sheltered places. Some adults are remarkably drought resistant and can survive for several months or years in a semidesert environment.

In tropical climates, the dominant factor is rainfall (Pegram et al., 1989). The start and end of the rainy season influence the different phases of the life cycle. Parasitism is reduced during the dry months.

Evaporation, transpiration and evapotranspiration

Evaporation

Evaporation is the process whereby liquid water is converted to water vapour and removed from the evaporating surface. Water evaporates from a variety of surfaces, such as oceans, lakes, rivers, soils and wet vegetation. Evaporation from big water bodies, such as oceans or lakes is essential to produce the water vapour that will lead to rain. Evaporation from storage dams that are built to provide water for irrigation and other uses is a negative factor. Dams that have big surface areas and/or are shallow are relatively inefficient, especially if they are in hot, dry areas.

In addition to solar radiation, which is usually not measured, the following climatologically factors determine evaporation rates:

- Temperature: Higher temperatures increase evaporation.
- Humidity: Drier surrounding air increases evaporation.
- Wind: Winds have different effects on evaporation. These include:

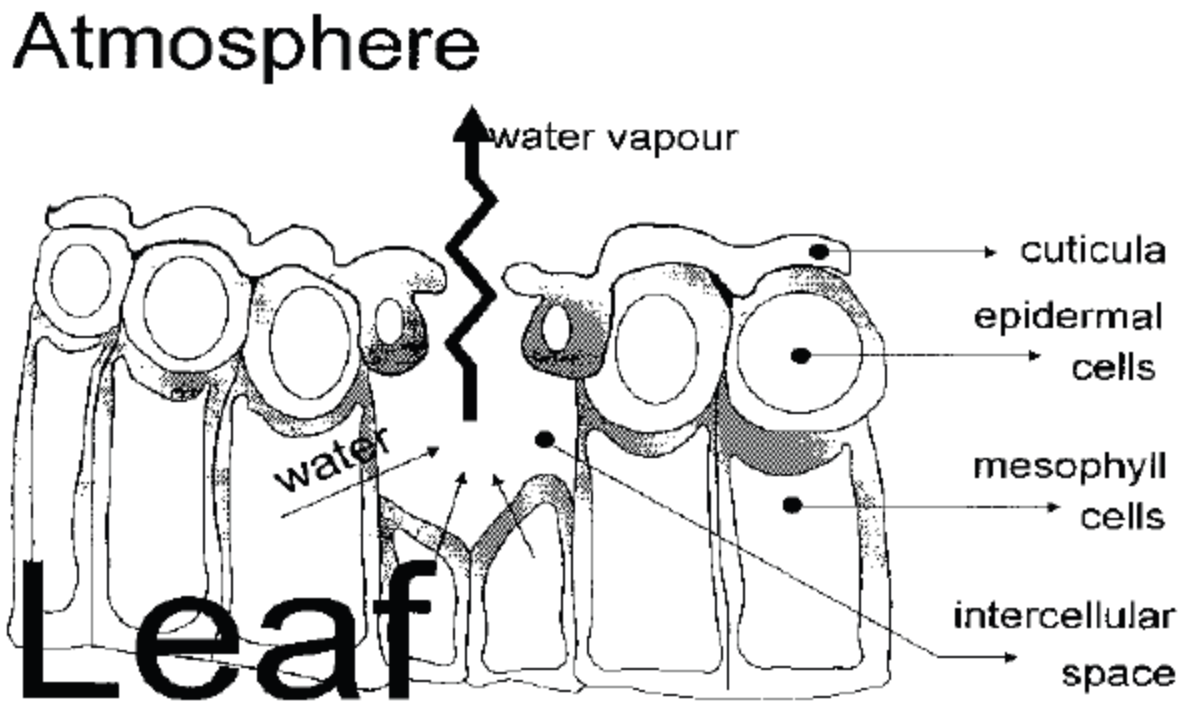
  A hot, dry wind entering an irrigated area will replace moist air with dry air and remove the moist air. This results in faster evaporation. This is an important factor in the central irrigation schemes

  Wind causes turbulence which mixes dry atmospheric air with the saturated air in the immediate vicinity of the evaporation surface. This results in faster evaporation.

  Cool, moist winds from a cold ocean or big delta increase humidities and reduce evaporation rates.

Transpiration

Transpiration consists of the vaporisation of liquid water contained in plant tissue and removal of the vapour to the atmosphere. Crops predominantly lose their water through stomata. These are small openings on the plant leaf through which gasses and water vapour pass. The water together with some nutrients is taken up by the roots and transported through the plant. Vaporisation occurs within the leaf, namely in the intercellular spaces, and the vapour exchange with the atmosphere is controlled by the stomatal aperture (size). Nearly all water taken up is lost through transpiration and only a tiny fraction is used within the plant.



Transpiration, like direct evaporation, depends on the energy supply, water vapour pressure gradient and wind. Thus, there is a direct relationship between the total evaporation experienced in an area and the total transpiration than can be expected in the same area.

The following factors affect transpiration rate:

a. Climatological factors: Air temperature, air humidity and wind affects transpiration in the same ways as discussed for evaporation.
b. Soil factors: Soil factors that affect transpiration rate include: Soil water content: The lower the soil water content, the slower is water supply to the plant roots until a point is reached where it becomes too slow to maintain a high transpiration rate. Pore characteristics of the soil: In some clayey soils micro-pores have such small diameters that they limit the rate at which water can be supplied to plant roots and thus limit the transpiration rate, even in a soil that is close to field capacity. Water logging: Water uptake by plant roots is an active process which means that root respiration is required for efficient water uptake. Under waterlogged conditions this is not possible and transpiration is restricted. Salinity: In saline soils, e.g. soils with high salt contents, a high osmotic tension develop which hampers water uptake by plants and reduces the transpiration rate.

Crop factors: Transpiration rate is also influenced by crop characteristics and the responses of crops to environmental conditions. Different types of crops have different transpiration rates. Not

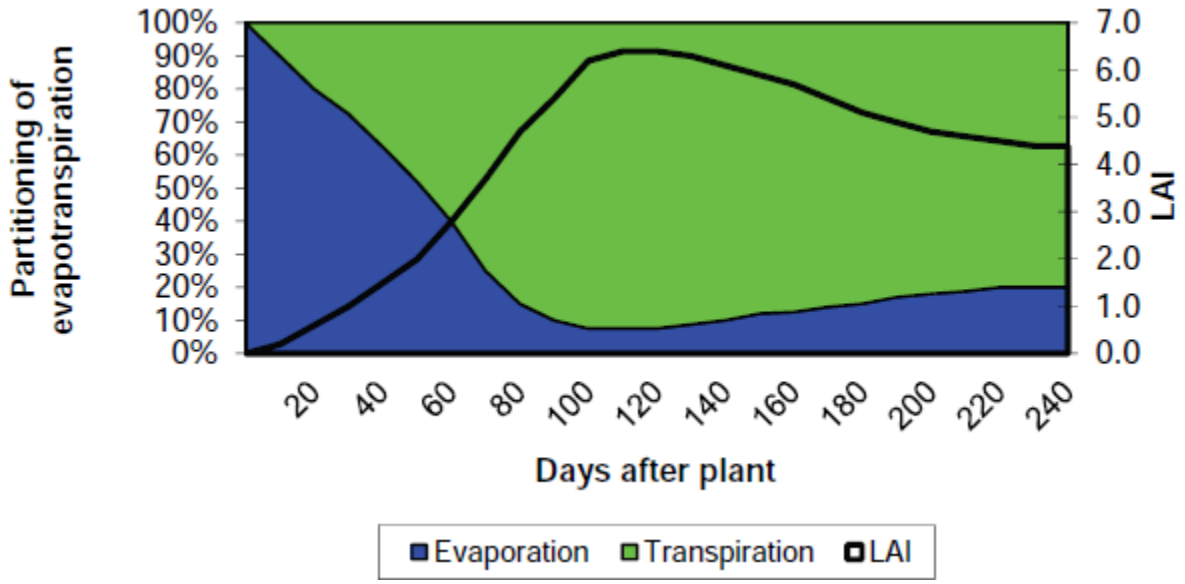
only the type of crop, but also the phonological (development) stage of the crop makes a difference. There are often also large differences between different cultivars or clones of the same crop. Some crop factors include:

a. Type of crop: For example, xerophytic (drought tolerant) plants have mechanisms that reduce transpiration rates, such as stomata that only open during the night and/or layers of hair on leaf surfaces that shade the leaves and reduce the effect of sunlight and wind at the leaf surface and/or or stomata only on the underside of the leaf.
b. Stomatal control: Some crops have the ability to exercise stomatal control, i.e. to close stomata, under extreme water stress conditions. This may include very high atmospherically induced transpiration demand conditions, caused by very hot, dry atmospheric conditions. Actual transpiration rates will then be much lower than the potential rates calculated. In some crops there are large differences between different cultivars or clones regarding their ability to exercise stomatal control.
c. Radial and axial resistances to water flow: Radial (sideways) resistances in roots and axial resistances in the vascular systems of roots, stems and shoots limit water transport rates in plants. Thus, actual transpiration rates may be much lower than potential rates as a result of this factor. Again, it will be more pronounced where the transpiration demand is very high.
d. Crop diseases: Corp diseases cause stress in the plant which has a tendency to reduce the transpiration rate. Root rot diseases reduce the ability of roots to absorb water and thus decrease the transpiration rate.
e. Plants in net houses transpire less than plants outside because of the different micro-climate caused by the protection against wind and direct sunlight as well as the higher humidity caused by the net cover.

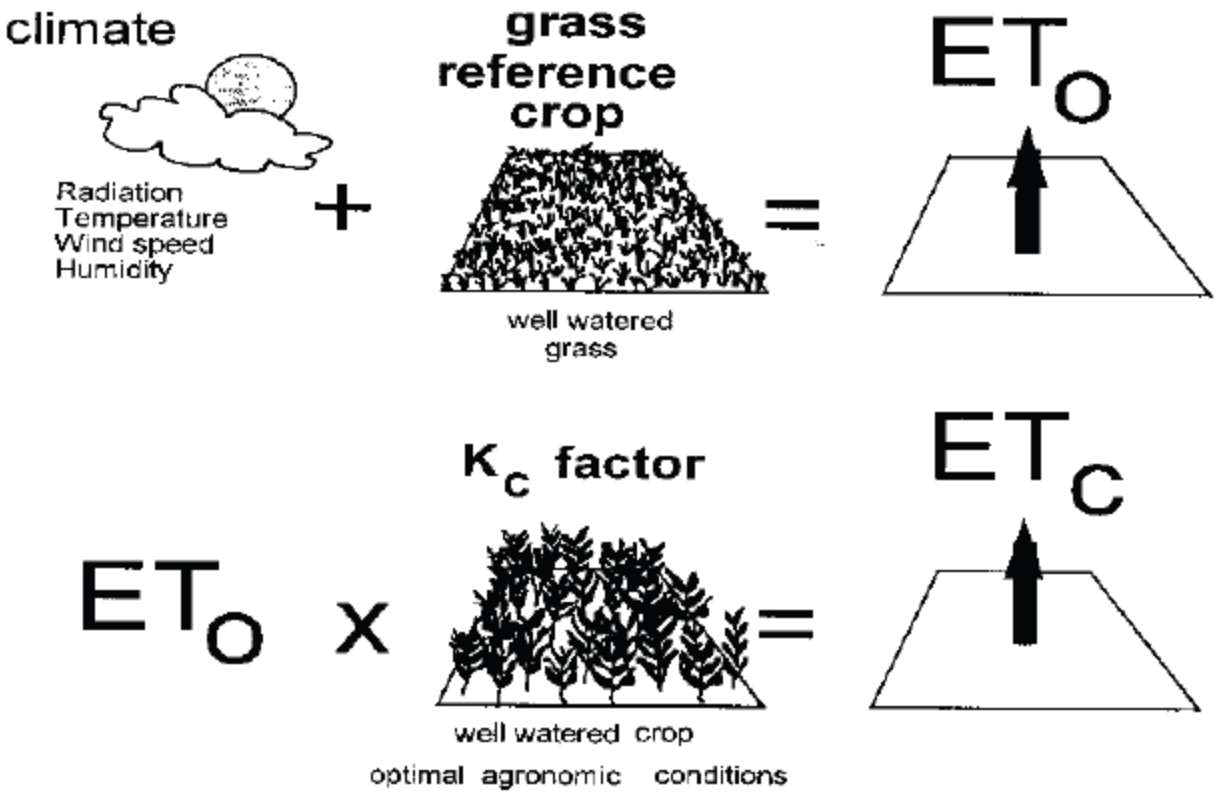
Evapotranspiration

Evaporation and transpiration occur at the same time and there is no easy way to distinguish between the two processes. Apart from the water availability in the topsoil the evaporation from a cropped soil is mainly determined by the fraction of the solar radiation reaching the soil surface. This fraction decreases over the growing season as the crop develops and the crop

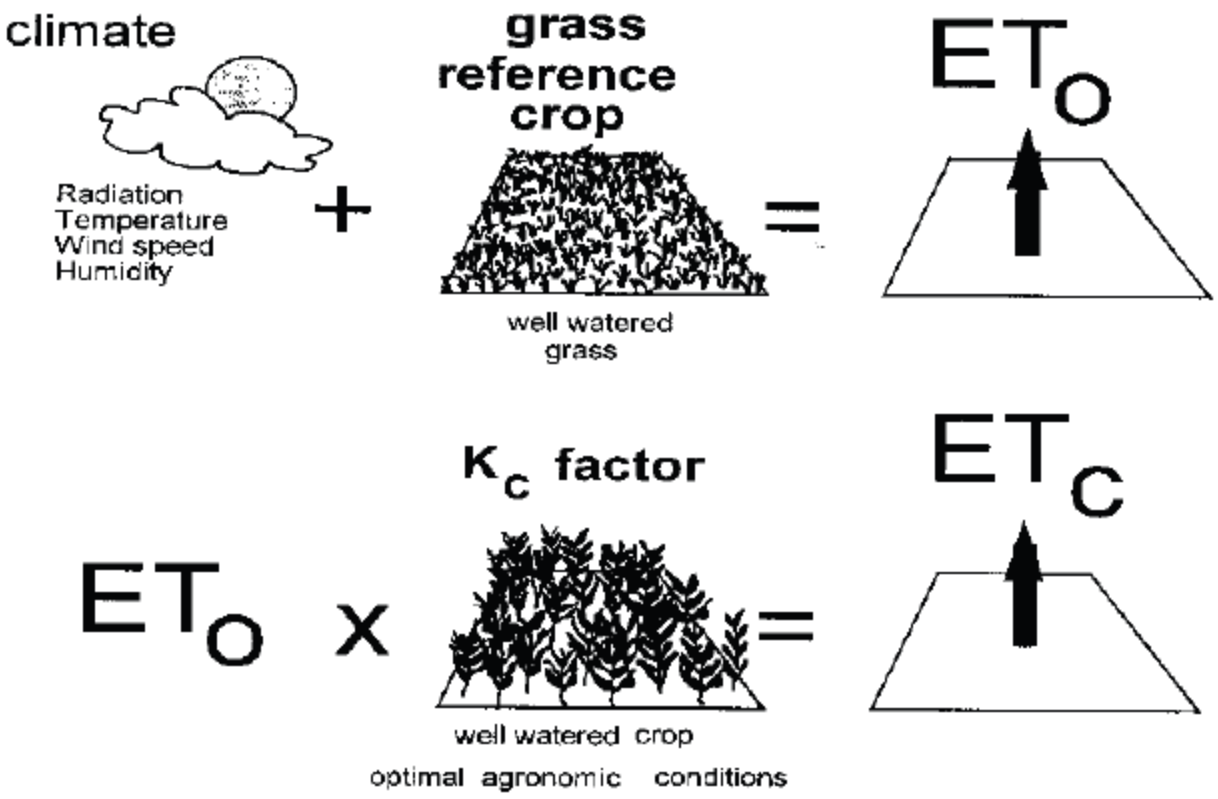
canopy shades more and more of the ground surface. When the crop is small, water is predominantly lost by evaporation from the soil, but as the crop develops, transpiration becomes the main process because more and more of the soil surface is shaded by foliage and total leaf area increases. At sowing nearly all of evapotranspiration is evaporation from the soil surface, but at full foliage cover more than 90% of evapotranspiration comes from transpiration .



Potential evapotranspiration is the evapotranspiration rate from a hypothetical grass reference surface with specific characteristics and never short of water. It is used as a standard against which the water requirement of all other crops is measured.



The concept of reference evapotranspiration was introduced to study the evaporative demand of the atmosphere independently of crop type, crop development and management practices



Wind

Global wind belts are found at certain latitudes north and south of the equator. Of interest to Zambia are the so-called trade wind belts of the mid-latitudes south and north of the equator. The trade winds develop as follows: Due to intense heating at the equator air rises. At great height this air moves away from the equator in the troposphere. At about 30o latitude in both hemispheres this air starts subsiding towards the surface in subtropical high-pressure belts called subtropical ridges. As the air descends, its temperature increases, but its absolute humidity remains the same. Consequently its relative humidity decreases. Thus around 30o a high pressure zone of stable dry air prevails. The result is a region with high temperatures and low rainfall.

The rising of the air at the equator causes low pressure at low levels in that zone. Air from high pressure trade wind zones moves towards the lower pressure zones. This air movement is not directly northwards (in the southern hemisphere) or southwards (in the northern hemisphere). It is deflected due to the rotation of the earth around its axis. The deflection is such that the winds

blow predominantly from the southeast in the southern hemisphere (known as the south-easterly trade winds) and from the northeast in the northern hemisphere.

Wind effects important for irrigated agriculture.

Wind effects on crops include:

- Gentle winds may benefit crops by replenishing CO2 to leaves in densely planted crops.
- Crop yields can be impaired by strong winds at critical times in the crop's production cycle, such as inhibiting bee activity during blooming period when it is essential for pollination.
- Some fruits are tender during certain development stages and whipping winds can cause skin blemishes, making the fruit unacceptable for marketing.
- Deciduous tree branches, heavily laden with ripening fruit can be broken by strong winds.
- "Tree throw" (i.e. blowing over) of mature fruit trees can happen.
- Trees with weak bud unions between the rootstock and scion can be snapped at the bud union by wind. Some rootstock/scion combinations have inherently weak bud unions. It sometimes means that a good cultivar on an excellent rootstock cannot be planted in a specific area due to wind hazard.
- Continuous winds accompanied by high temperatures during the growing season markedly increases water losses through transpiration and evaporation, to such an extent that the plant may not be able to cope with it.
- Increased air flow through the canopy of densely planted crops can reduce disease incidence and improve photosynthetic efficiency by refreshing CO2 at the surface of the leaves.
- Crops can be "sandblasted" or even buried under sand where wind erosion is a problem
- Vineyards receiving cool sea breezes may produce better quality wines.

Wind effects on overhead irrigation systems

Strong winds can disturb the water distribution patterns of overhead irrigation systems and reduce their efficiencies. In relatively moderate cases system design can be adapted to compensate for this. In extreme cases an area can be unsuitable for the use of such systems. An

example of the latter was found in the panhandle of Texas: Farmers traditionally used furrow irrigation. In the 1970s they switched to overhead sprinkler irrigation. The effects of the high winds prevailing in the area on the systems were so severe that by the early 1980s the farmers were forced to switched back to furrow irrigation. Hot, dry winds can also cause very high evaporation losses during water application by such systems.

Windbreaks

Where soil and all other climatic conditions are suitable for irrigated crop production, but a serious wind hazard prevails, windbreaks are used to overcome the problem. Normally tall growing trees are planted as windbreaks, but other types of windbreaks can also be effective for certain situations.

Trees as windbreaks

The following publication by the KwaZulu-Natal Department of Agriculture gives an excellent outline of the design, planning, establishment and maintenance of tree windbreaks. It also gives information on the selection of suitable trees for different ecological regions in summer rainfall areas, as well as a comprehensive list of tree species that can be used. It also outlines how windbreaks work and the advantages and disadvantages of windbreaks, some of which are listed below.

a. Some of the main advantages of trees as windbreaks are:

• Wind speed is greatly reduced, thus

- Preventing/limiting wind throws of trees, breaking of fruit-laden branches, etc.
- Preventing physical damage to flowers, fruit and plants by dust, hail, snow andabrasion. Both the quantity and the quality of the crop is improved.
- Facilitating pollination by bees, which is especially important for fruit production.
- Windbreaks can supply products such as timber, poles and firewood, as well as nectar and pollen for honey production.
- Windbreaks can provide shelter for desirable birds and animals.

*Some of the main **disadvantages** of trees as windbreaks are:*

- Windbreaks can have negative impacts on the micro-climate in an orchard or field. For example: Windbreaks on the contour can trap cold air (resulting in frost pockets) or hot air (resulting in heat scorch), both of which can cause crop damage. Normally evergreen

trees are used as windbreaks. However, where protection is required against summer winds and there is a danger of frost pockets in winter, it may be preferable to plant deciduous windbreaks. A reduction in windbreak density in deciduous windbreaks during winter, due to being leafless during that period, will allow some air drainage through the windbreak and reduce the formation of frost pockets in orchards.

- A too dense windbreak barrier can cause winds to become turbulent of the leeward side (i.e. in the orchard or field), causing physical damage to crops.
- Undesirable birds may find shelter in windbreaks.
- If a wrong tree species is chosen it may invade adjoining areas.
- Some windbreak tree species may be hosts for the survival of insect pests and/or disease organisms affecting the crop.